C37$_A$3

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1 Q1

1.1 a

1.1.1 Convergence Test

1. Residual Convergence Test

- we can define convergence in terms of the residual $r^{(k)} = b - A\tilde{x}^{(k)}$

- if the norm of the residual $||r^{(k)}||$ is below a predefined tolerance error ε_r, we can consider the solution sufficiently accurate.

- The test is $||r^{(k)}|| < \varepsilon_r$

- This test essentially checks if $\hat{x}^{(k)}$ satisfies the equation $Ax = b$ within the tolerance ε_r.

2. Solution Update Test

- Another way to define convergence is based on the magnitude of the update in each iteration.

- If the norm the update $||\hat{z}^{(k)}||$ becomes very small, which means we can recognized the algo is converged.

- The test is $\|\hat{z}^{(k)}\| < \varepsilon_x$

- Here, the ε_x is a small tolerance value, typically chosen to be close to machine precision.

1.1.2 Convergence Failure Tests

1. Maximum Iteration test

- If the IR algo does not converge within a set maximum number of iteration k_{max}, we can consider it a failure to converge

- This prevent the algo on infinity loop

2. Condition Number Test

- The condition number $\|(A)$ of matrix A indicate how sensitive the matrix is to numerical errors and can affect the convergence of the IR algo.

- If the condition number very large, the matrix A is ill-conditioned, and iterative refinement may not converge.

- This test is $\|(A) > Threshold$, which Threshold could be a large value such as 10^8 based on numerical stability requirements.

1.2 b

According to the definition of the "update ... ", we get $\|\hat{x}^{(k+1)} - x\| = \|\hat{x}^{(k)} + \hat{z}^{(k)} - x\|$.

According to the Residual System, we have $r^{(k)} = b - A\hat{x}^{(k)}$, and we can get the $\hat{z}^{(k)}$ by $Az^{(k)} = r^{(k)}$. However, due to the round-off errors, we need to get the answer by $(A+E)\hat{z}^{(k)} = r^{(k)}$.

Let $z^{(k)}$ be the exact solution of $Az^{(k)} = r^{(k)}$ and $\delta z^{(k)} = \hat{z}^{(k)} - z^{(k)}$ be the error in exact solution due to the E.

According the above terms, we can get the $A\delta z^{(k)} = -E\hat{z}^{(k)}$. In this term, we can know that the error $\|\delta z^{(k)}\|$ depends on the condition number of A and the E.

Thus, we have $\|\hat{x}^{(k+1)} - x\| = \|\hat{x}^{(k)} + \hat{z}^{(k)} - x\| \approx \|\hat{x}^{(k)} - x\| + \|\delta z^{(k)}\|$, where $\|\delta z^{(k)}\|$ is proportional to $\|E\| \cdot \|(A) \cdot \|\hat{x}^{(k)} - x\|$ with $\|(A)$ being the condition number of A.

For Converge, we required that the error doesn't grow with each iteration. If A is well-conditioned, then the error remain small, and thus the amplification is limited. This ensures that the sequence $\|\hat{x}^{(k+1)} - x\|$ converges to zero as $k \to \infty$. Otherwise, if A is poor-conditioned, then amplification factor will lead the error grow, which cause the divergence.

2 Q2

There isn't an algebra equivalent. Because when we doing the $\frac{b_i^2-(\sum_{j=1}^{n} a_{ij}\hat{x}_j^{(k)})^2}{b_i+\sum_{j=1}^{n} a_{ij}\hat{x}_j^{(k)}}$ is also face the same subtraction problem.

First of all, the residual itself is a subtraction structure. There is no direct algebraic manipulation to avoid the subtraction cancellation. Here is some possible advice to minimize the effect:

1. We can evaluate $\sum_{j=1}^{n} a_{ij}\hat{x}_j^{(k)}$ as much as possible before it calculates with b_i. When we are doing the calculation, we can use some methods to keep the precision, for example, Kahan summation or others.

2. If possible, using the higher precision than the standard floating-point precision (double-precision or quad precision)

3. Using the compensated summation techniques, such as the Kahan summation algo, when calculate the summation of $\sum_{j=1}^{n} a_{ij}\hat{x}_j^{(k)}$ will reduce the floating error.

4.Rescaling the parameter, By scaling A, x, and b so that their entries are within a comparable range of magnitudes, you may reduce the occurrence of subtractive cancellation.

5. Checking the residual in each iteration. Monitoring the size of the residual and stop the IR iterations if the residual does not decrease as expected. If the residual stagnates or increases, it could indicate that the error, which produced by subtractive cancellation or other numerical issues.

3 Q3

Algorithm 1 IR Algorithm with LU Factorization

1: Compute $PA = LU$ ($O(n^3)$ operations)
2: Apply permutation P to b: Pb
3: Solve $Ly = Pb$ using forward substitution ($O(n^2)$ operations)
4: Solve $U\hat{x}^{(0)} = y$ using back substitution ($O(n^2)$ operations)
5: **for** $k = 0, 1, 2, \ldots, M-1$ **do**
6:     Compute the residual: $r^{(k)} = b - A\hat{x}^{(k)}$ ($O(n^2)$ operations)
7:     Solve $Az^{(k)} = r^{(k)}$ using LU decomposition:
8:         Apply permutation to $r^{(k)}$: $Pr^{(k)}$
9:         Solve $Ly = Pr^{(k)}$ using forward substitution ($O(n^2)$ operations)
10:         Solve $U\hat{z}^{(k)} = y$ using back substitution ($O(n^2)$ operations)
11:     Update the solution: $\hat{x}^{(k+1)} = \hat{x}^{(k)} + \hat{z}^{(k)}$ ($O(n)$ operations)
12: **end for**

Complexity Analyze For Compute the LU Factorization, we cost total $O(n^3)$. Line 2 cost $O(1)$. Then, we solve the initial solution, which backward and frontward total cost $O(n^2)$. After that we start our loop iteration, assume it converge to M times. Residual computation will cost $O(n^2)$ and Solve the $Az^{(k)} = r^{(k)}$ will total cost $O(n^2) + O(n) = O(n^2)$, the iteration total costs $M \cdot O(n^2) = O(M \cdot n^2)$.

For conclution, the total cost is that $O(n^3) + O(n^2) + O(M \cdot n^2) = O(n^3 + M \cdot n^2)$

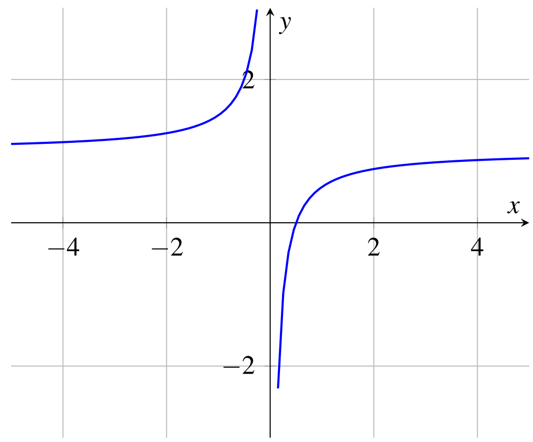

Figure 1: Plot of $f(x) = 1 - \frac{1}{2x}$

4 Q4

4.1 a

For f is a Inverse proportional function, there is only one root $x = \frac{1}{2}$ when $f(x) = 0$.

Because, when $x = \frac{1}{2}$, $g(x) = 2 \times \frac{1}{2}(1 - \frac{1}{2}) = \frac{1}{2}$, we say that the root of $f(x)$ is the fixed point of $g(x)$.

For we want to find the fixed point, as same as we want to find the x of $x = g(x) = 2x(1-x)$, where we will get $x = 2x - 2x^2 \Rightarrow 0 = x - 2x^2 \Rightarrow 0 = x(1-2x)$, then we will get $x_1 = 0$ and $x_2 = \frac{1}{2}$, which are the fixed points of $g(x)$. Then we conclude there are more fixed points of g than roots of f.

4.2 b

From the part a, we know the root of $x = \frac{1}{2}$ is the root of $f(x)$

we want to get the second form $g(x) = x - h(x)f(x)$, from the question, we substitute the $g(x)$ and $f(x)$.

Then we get $2x - 2x^2 = x - h(x)(1 - \frac{1}{2x}) \Rightarrow h(x) = \frac{2x^2 - x}{1 - \frac{1}{2x}} = \frac{2x(2x^2 - x)}{2x - 1} = 2x^2$

4.3 c

For we know that $g(x) = 2x(1-x) = 2x - 2x^2$, then we can get $g'(x) = 2 - 4x$. By FPT, we want to find an interval $[a,b]$ such that $g(x) \in [a,b]$ for all $x \in [a,b]$.

For meet the condition, we calculate the $|g'(x)| \leq L < 1$ first.

then we get $-1 < 2 - 4x < 1$. After doing the operation of inequality, we get $\frac{1}{4} < x < \frac{3}{4}$

Now, we verify this interval whether meet the mapping condition. for all $x \in (\frac{1}{4}, \frac{3}{4})$, we refactor the $g(x) = 2x - 2x^2 = 2(-x^2 + x) = -2(x^2 - x) = -2((x - \frac{1}{2})^2 - \frac{1}{4})$. Follow by this structure, we can know the interval of $g(x) \in (\frac{3}{8}, \frac{1}{2}]$, when $x = \frac{1}{4}$ or $x = \frac{3}{4}$, and $x = \frac{1}{2}$, which contained in the origin group.

The largest possible interval for which the fixed-point iteration $x_{k+1} = 2x_k(1 - x_k)$ is guaranteed to converge, according to the Fixed Point Theorem, is: $\left(\frac{1}{4}, \frac{3}{4}\right)$.

Additional Analysis Without additional information or a different approach beyond the Fixed Point Theorem, this interval $\left(\frac{1}{4}, \frac{3}{4}\right)$ is likely the maximum guaranteed interval of convergence based solely on the FPT, as it fulfills both the contraction and self-mapping conditions. Further analysis or more advanced techniques might be able to identify a larger interval, but this goes beyond the Fixed Point Theorem itself.

5 Q5

First we should determined the fixed point $x = e^{-x}$, the approximate value is $x = 0.5671$.

Then we apply the FPT. According to the $g(x) = e^{-x}$, we can get that $g'(x) = -e^{-x}$. Since the e^{-x} is always positive, the $g'(x)$ must be negative. We only need to solve that $-e^{-x} > -1$, which is as same as $e^{-x} < 1$, we get $x \in (0, +\infty)$

Now we need to map $[a,b]$ back into $[a,b]$. From the previous step we get the interval $(0, +\infty)$. Let's start by checking if there's a maximum interval $[0,b]$ over which $g(x) \in [0,b]$.

For $g(x) = e^{-x}$ is monotonically decreasing on $\mathscr{R}$. Thus, we calculate the $g(x) = 1$ when $x = 0$ and $g(x) = 0$ when $x \to +\infty$. Thus, $g(x) = e^{-x}$ is bounded within $(0,1]$, since e^{-x} approaches 0 as $x \to \infty$ and is never greater than 1 for $x \geq 0$.

The maximum interval [0, 1] satisfies both conditions:

- $g(x) = e^{-x} \in [0,1]$ for all $x \in [0,1]$.

- $|g\prime(x)| = e^{-x} < 1$ for all $x \in [0,1]$.

Therefore, the maximum guaranteed interval of convergence for the iteration $x_{k+1} = e^{-x_k}$ is: $[0,1]$.